Electron-Ion Path Integral Monte Carlo

Riccardo Fantoni*
Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy
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I. INTRODUCTION

PIMC with $M = \beta/\tau$ the number of timeslices of length τ.

Two-Component-Plasma (TCP) with masses $m_1 = m$ and $m_2 = 10^3 m$. N_i is the number of particles of species i and $N = \sum_i N_i$ with $x_i = N_i/N$ the molar fractions such that $\sum_i x_i = 1$.

$$\phi_{ij}(r) = \begin{cases} \epsilon_{ij}/r & r > \Sigma_{ij} \\ \infty & \text{else} \end{cases}, \tag{1.1}$$

where

$$\boldsymbol{\epsilon} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \tag{1.2}$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} 0 & \sigma/2 \\ \sigma/2 & \sigma \end{pmatrix}. \tag{1.3}$$

From Figure 2 we see the formation of clusters made of an heavy positive particle (of species 2) surrounded by a cloud of light negative particles (of species 1). We also see how changing the statistics between particles of the same species from Bose-Einstein to Fermi-Dirac makes them dislike themselves more as expected from the Pauli exclusion principle.

Note that at a given T if we choose σ too small the whole system collapses with all the negative light particles falling on the heavy positive particles.

Metallic Hydrogen

Let us consider a mixture of electrons, e, and protons, p. In reduced units we will consider $\hbar = k_B = 1$. We measure masses in units of the electron mass m_e so that the reduced mass of the electron will be $\mu_e = m_e/m_e = 1$ and the reduced mass of the proton $\mu_p = m_p/m_e \approx 1836.15$. The characteristic kinetic energy is

$$\mathscr{K} = \frac{\hbar^2}{2m_e\ell^2}, \tag{1.4}$$

* riccardo.fantoni@scuola.istruzione.it

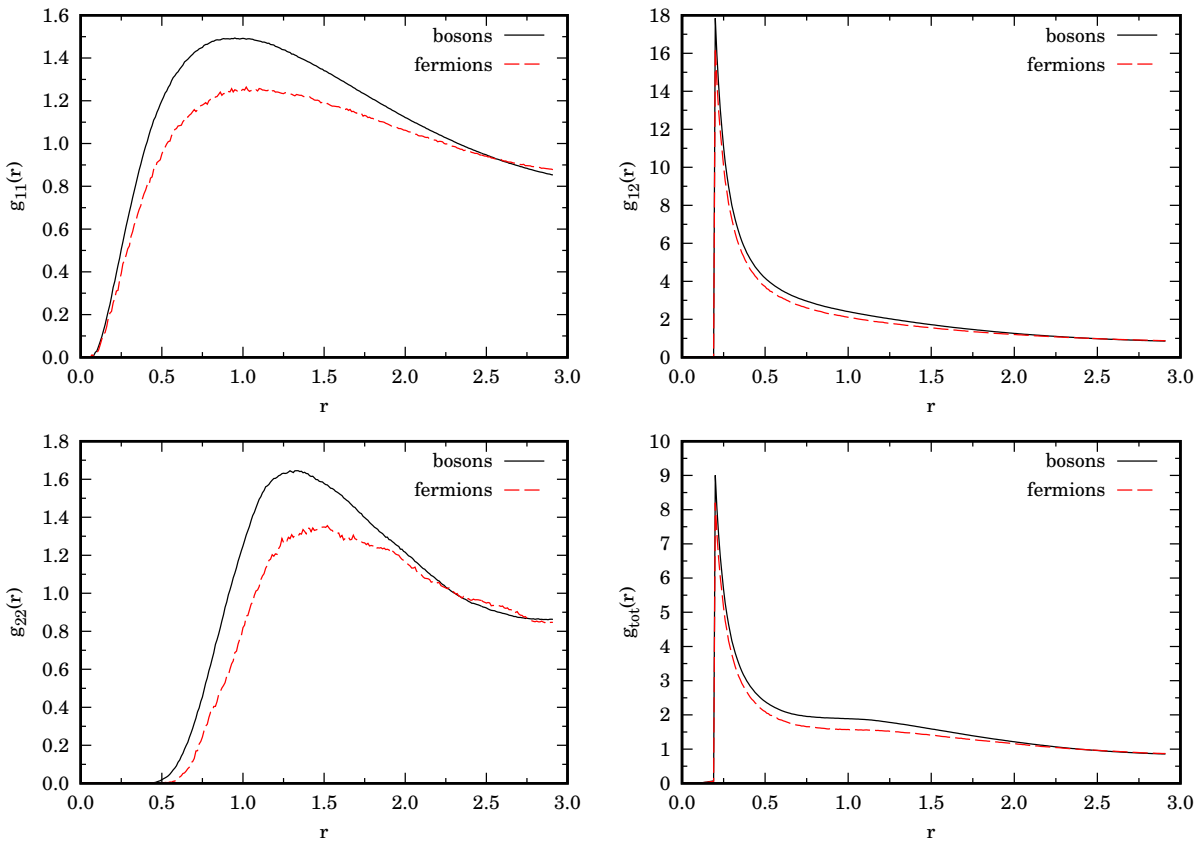

FIG. 1. The partial, g_{ij}, and total, g_{tot}, radial distribution functions for a binary mixture with $N = 20, x_1 = 1/2, m_1 = 1, m_2 = 1000, \sigma = 0.4$. The mixture is in a thermodynamic state with a reduced temperature $T = T'/\mathscr{T} = 0.1$ with T' measured in degrees Kelvin and a reduced number density $n = n'\mathscr{L}^3 = 0.1$ with n' measured in cm^{-3}. In the path integral we chose $M = 20$ and used either Bose-Einstein or Fermi-Dirac statistics among like particles. In the figures $r = r'/\mathscr{L}$ where r' is measured in centimeters.

with ℓ a length. The characteristic Coulomb energy is

$$\mathscr{V} = \frac{e^2}{\ell} = \frac{2\mathscr{K}\ell}{\mathscr{L}}, \tag{1.5}$$

with e the electron charge magnitude and $\mathscr{L} \approx 5.29177 \times 10^{-9}$cm a characteristic length. Then, if we denote with n the reduced number density this will correspond to a real number density of $n\mathscr{L}^{-3}$. For example for metallic hydrogen we have a mass density $\approx 1\mathrm{g/cm}^3$ which corresponds to a reduced density of $n \approx \mathscr{L}^3/m_p \approx 0.0885943$.

The size of the nucleus of the hydrogen atom is $\approx 2 \times 10^{-13}$cm which in reduced units is $\sigma \approx 3.77945 \times 10^{-5}$.

We may then take as characteristic temperature

$$\mathscr{T} = \frac{\mathscr{K}\ell^2}{\mathscr{L}^2 k_B} \approx 315775\mathrm{K}. \tag{1.6}$$

So that for metallic hydrogen in planetary interiors an absolute temperature of $\approx 10^5$K will correspond to a reduced temperature of $T \approx 10^5/\mathscr{T} \approx 0.0316681$.

The Hamiltonian and the density matrix in reduced units will then read

$$H = -\sum_{i_e} \frac{1}{2\mu_e}\nabla^2_{i_e} - \sum_{i_e} \frac{1}{2\mu_p}\nabla^2_{i_p} + \sum_{i_e<j_e} \phi_{i_e j_e}(r_{i_e j_e}) + \sum_{i_p<j_p} \phi_{i_p j_p}(r_{i_p j_p}) + \sum_{i_e i_p} \phi_{i_e i_p}(r_{i_e i_p}), \tag{1.7}$$

$$\rho = e^{-H/T}. \tag{1.8}$$

with ϕ_{ij} given by Eq. (1.1) and distances are measured in units of $\mathscr{L}$.

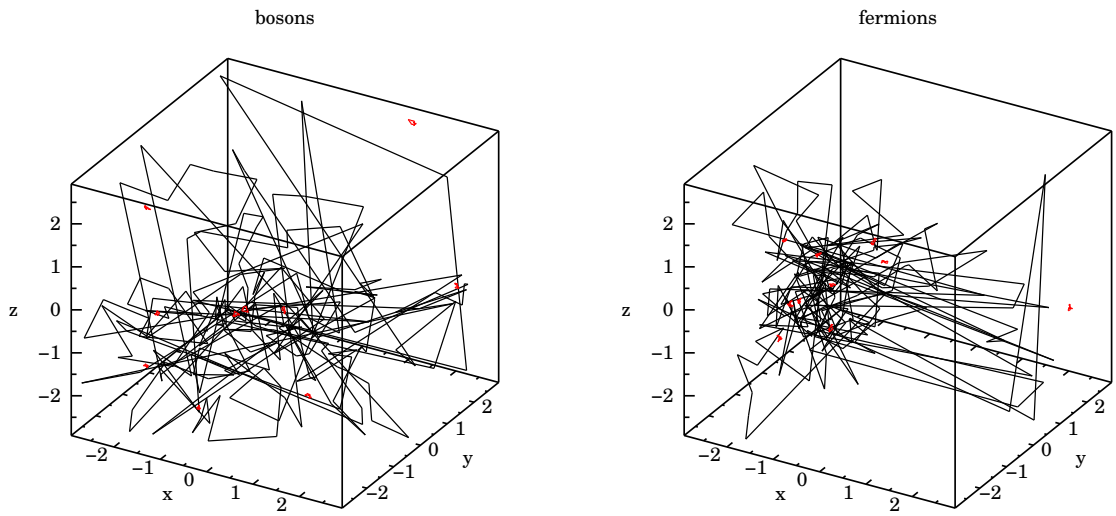

FIG. 2. Snapshots of the simulation of a binary mixture with $N = 20, x_1 = 1/2, m_1 = 1, m_2 = 1000, \sigma = 0.4$. The mixture is in a thermodynamic state with a reduced temperature $T = T'/\mathscr{T} = 0.1$ with T' measured in degrees Kelvin and a reduced number density $n = n'\mathscr{L}^3 = 0.1$ with n' measured in cm^{-3}. In the path integral we chose $M = 20$ and used either Bose-Einstein (left panel) or Fermi-Dirac (right panel) statistics among like particles. Red paths are for the heavy slow species and black paths are for the light fast species.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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